

WEAVE: Wavelet-Decoupled Velocity Matching for Lightweight Latent Style Transfer

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Abstract—Diffusion-based style transfer produces strong stylization but exceeds the memory budget of consumer GPUs, while many lightweight alternatives drift toward an identity mapping that scores well on content metrics yet adds little style. We trace this failure to frequency dominance: in a shared latent objective, large low-frequency structural residuals overwhelm the high-frequency style signal. We propose WEAVE, a wavelet-decoupled velocity-matching framework in the latent space of a frozen VAE. A one-level Haar transform separates structure from oriented texture, a compact velocity network learns source-aligned transport with a down-weighted low-frequency loss, and a stepwise statistical alignment injects target style into the high-frequency bands at every integration step. We also report identity and target-style references that expose style suppression and content collapse. With 1.04M trainable parameters and 1.4 minutes of training on one RTX 3060, WEAVE exceeds the identity reference on three benchmarks while preserving more content than diffusion editors.

Index Terms—style transfer, flow matching, wavelet decomposition, latent space, efficient generation

I. INTRODUCTION

Image foundation models such as Qwen-Image and FLUX.2 provide strong editing priors but can require more than 60 GB of GPU memory [1], [2]. Lightweight style-transfer models avoid this cost, yet some preserve the source so strongly that their style affinity falls *below the unchanged input*. Standard protocols hide this no-op behavior: LPIPS rewards source reconstruction [3], the content factor of ArtFID can favor a near-copy [4], and the identity mapping is rarely reported as a baseline.

Fig. 1 makes the missing comparison explicit with two trivial outputs. Identity (IDT) returns the source; Target Style (TGT) returns an unrelated image of the requested style. A useful transfer should move beyond IDT in style without moving as far from the source as TGT in content. Methods below IDT have not acquired the target style; methods beyond TGT have discarded as much content as a source-independent style image. We call this bracket the IDT–TGT sandwich and report it alongside DINO, CLIP, and LPIPS scores.

Why do compact models enter the identity regime? Networks fit low frequencies first [5], [6], and natural images concentrate energy in layout rather than brushstroke. A velocity field trained on the full latent must explain both with one output, so the larger low-frequency residuals dominate the gradient and weaken style-conditioned high-frequency motion. Our subband probes (Fig. 2) measure this imbalance directly; widening the network does not remove it because both signals still share one basis and one objective.

We therefore change the representation and the target before adding capacity. WEAVE caches latents of a frozen SD1.5 VAE [7] and splits them with a one-level Haar transform into structure (LL) and oriented texture (LH/HL/HH). A 1.04M-parameter velocity network learns source-aligned transport under a down-weighted LL loss; pooled target codes modulate only the LH/HL heads; and an AdaIN update aligns the high-frequency residual after every Euler step. Training takes 82.8 s on one RTX 3060 and stops at an internal gradient transition without any image evaluation.

Contributions. (1) An evaluation guideline that reports IDT and TGT references with every metric, exposing style suppression and content collapse in raw metric units. (2) A measurement of low-frequency gradient dominance in latent velocity matching and a wavelet-decoupled transport that assigns structure, learned texture motion, and trajectory-level statistics to explicit subbands. (3) A 1.04M-parameter model trained in 1.4 minutes that clears IDT on three benchmarks, with matched ablations, seed and reference-pool checks, and a training-free plug-in for a frozen SD1.5 editor.

II. RELATED WORK

Neural and statistical style transfer. Gatys et al. matched Gram statistics by optimization [8], [9]; AdaIN and WCT replaced it with feed-forward moment and covariance transforms [10], [11]. SANet, StyTR-2, and AesPA-Net add attention [12]–[14], CUT learns unpaired translation by patch contrast [15], and Mamba-based SaMST and SaMam reduce sequence cost [16]–[18]. These methods mix structure and appearance in one feature tensor; WEAVE restricts statistical correction to explicit frequency coordinates.

Diffusion editors. Latent diffusion [7] enables SDEdit, Prompt-to-Prompt, StyleID, StyleAligned, InST, Z-STAR, IP-Adapter, and StyleShot [19]–[26]. Being “training-free” leaves a 1B-parameter denoiser, often with inversion, in every call. WEAVE keeps only the frozen VAE; our plug-in experiment tests the Haar operator inside such an editor.

Flow matching and bridges. Rectified flow and flow matching [27], [28] regress velocities along endpoint interpolations, and Schrödinger-bridge solvers [29]–[32] address unpaired domains, but their objectives weigh all coordinates together. WEAVE changes the endpoint, basis, and per-band weights of the same regression.

Frequency-aware and wavelet methods. FDIT constrains low and high frequencies separately for translation [33], FreqFlow weights frequencies for class-conditional generation [34], and wavelets support diffusion [35] and photorealism

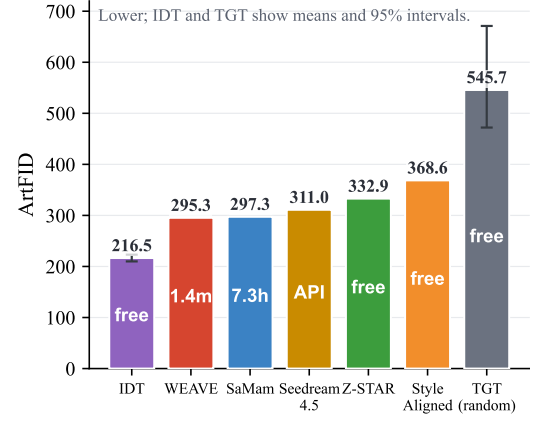
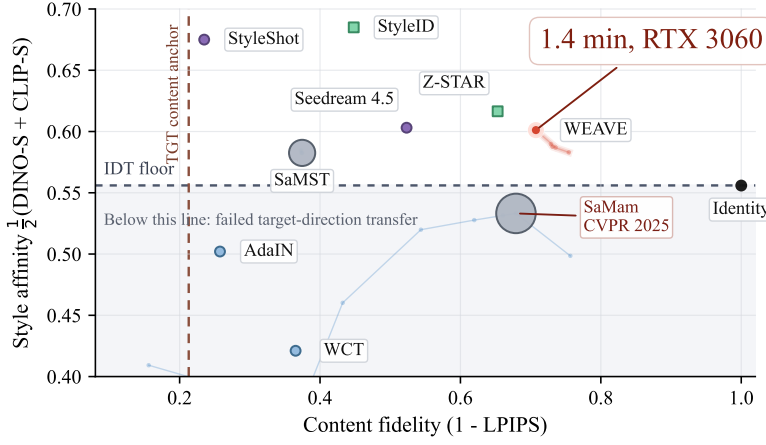


Fig. 1. IDT-TGT sandwich on D5-512. Left: style affinity versus content fidelity; bubble area is training time. Methods below the IDT line have not moved toward the target style; methods left of the TGT line have lost as much content as an unrelated target-style image. Right: target-pooled ArtFID, where IDT scores best, showing why a no-op reference is needed (IDT: 95% bootstrap interval; TGT: 2.5–97.5 percentile over 30 reference draws).

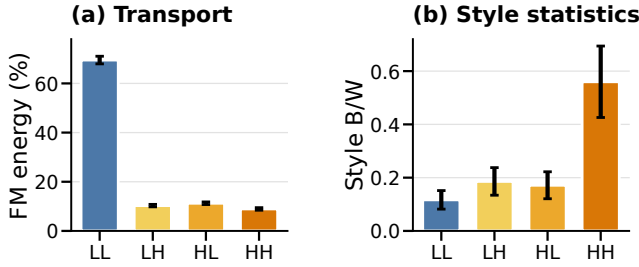


Fig. 2. Frequency probes on 5,000 cross-style latent pairs. (a) Share of endpoint-displacement energy per Haar band. (b) Between-/within-style variance ratio of per-image band statistics. Transport energy concentrates in LL while style separation is strongest at high frequency.

tic transfer [36]. WEAVE uses the orthogonal Haar basis [37] *inside* the transport objective, with a distinct role per band.

III. METHOD

A. Latent Haar Coordinates

Let $z_c = \mathcal{E}(x_c) \in \mathbb{R}^{4 \times H \times W}$ and $z_s = \mathcal{E}(x_s)$ be content and target-style latents of the frozen VAE. The one-level Haar transform $\mathcal{W}(z) = (\ell, h_1, h_2, h_3)$ is orthonormal and exactly invertible, so losses and heads can be assigned to explicit coordinates. We write $P_L z = \mathcal{W}^{-1}(\ell, 0, 0, 0)$ and $P_H = I - P_L$; latents with equal $P_L z$ share the LL coordinate but may differ in local appearance. Across 5,000 cross-style pairs, displacement energy concentrates in LL while style separability is largest in HH (Fig. 2): the band that dominates the regression is not the band that carries style. The separation is exact in latent space, although the nonlinear decoder can still couple visible structure and appearance.

B. Source-aligned Endpoint

A direct path from z_c to z_s also learns the unrelated geometry of x_s . WEAVE instead keeps the source LL as an

anchor and takes high frequencies from the target:

$$z_1^* = \mathcal{W}^{-1}((1-\alpha)\ell_c + \alpha \text{AdaIN}(\ell_c, \ell_s), h_{1,s}, h_{2,s}, h_{3,s}), \quad (1)$$

with $\alpha = 0.3$, which permits a limited palette shift. Training samples a lightly perturbed interpolation

$$z_t = (1-t)z_c + tz_1^* - \sigma\sqrt{t(1-t)}\epsilon, \quad u = z_1^* - z_c, \quad (2)$$

with $t \sim \mathcal{U}[0, 1]$, $\epsilon \sim \mathcal{N}(0, I)$, and $\sigma = 0.02$. The perturbation vanishes at both endpoints and is absent at inference; the label remains the clean displacement.

C. Band-weighted Velocity Matching

The network $v_\theta(z_t, t, s)$ predicts $(v_\ell, v_{h_1}, v_{h_2})$ with separate 1×1 heads on a shared trunk of four width-64 residual blocks. With $(u_\ell, u_{h_1}, u_{h_2}, u_{h_3}) = \mathcal{W}(u)$,

$$\mathcal{L} = \mathbb{E}_t [\lambda_{LL} \|v_\ell - u_\ell\|^2 + \|v_{h_1} - u_{h_1}\|^2 + \|v_{h_2} - u_{h_2}\|^2], \quad (3)$$

with $\lambda_{LL} = 0.3$. Changing the endpoint (Eq. 1) alters the *requested* velocity, whereas λ_{LL} alters its optimization pressure. A style memory indexed by the target-style id gives a coarse prior to the trunk. For image-specific texture, the target LH and HL bands are pooled into coordinate-free, orientation-specific codes that enter only the matching heads; raw spatial target maps would leak target layout. HH has no learned head. The complete model has 1,037,087 trainable parameters.

D. Stepwise High-frequency Alignment

Velocity matching organizes LL/LH/HL motion but leaves a weak appearance signal. For a residual $q = P_H z$ and target residual $q_s = P_H z_s$, let $\mathcal{A}(q; q_s) = \sigma(q_s)(q - \mu(q))/\sigma(q) + \mu(q_s)$ with channel-wise spatial statistics [10]. One of K inference steps is

$$\begin{aligned} \bar{z}^{k+1} &= z^k + \frac{1}{K} \mathcal{W}^{-1}(v_\ell^k, v_{h_1}^k, v_{h_2}^k, 0), \\ z^{k+1} &= P_L \bar{z}^{k+1} + (1 - \beta) q^{k+1} + \beta \mathcal{A}(q^{k+1}; (1 + \eta) q_s), \end{aligned} \quad (4)$$

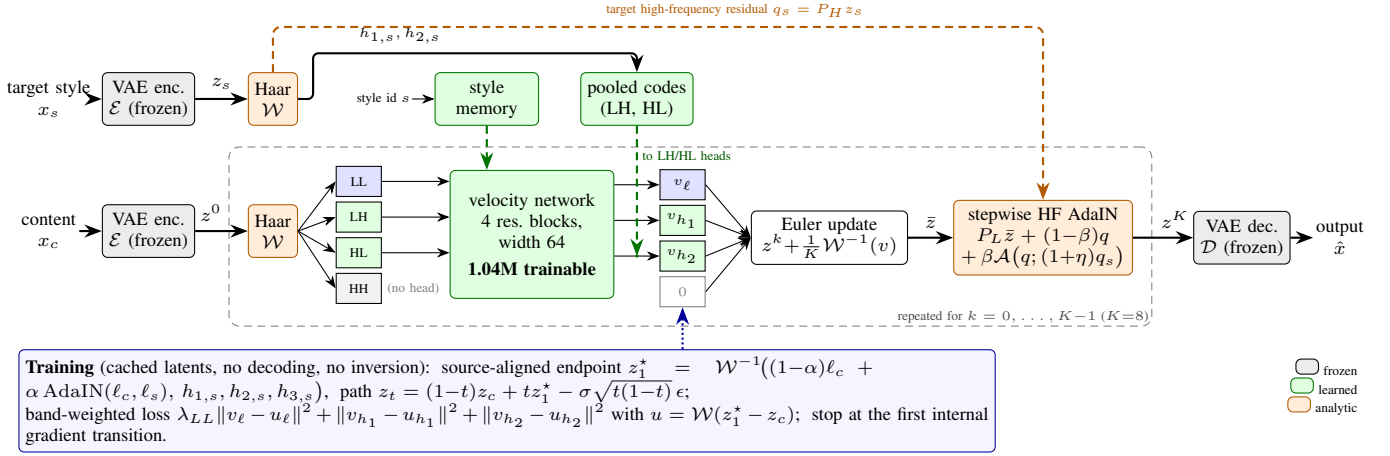


Fig. 3. WEAVE. A one-level Haar transform exposes LL and three high-frequency bands of the VAE latent. The compact network predicts LL/LH/HL velocities; pooled target LH/HL codes modulate only the matching heads and HH has no learned head. After every Euler update, AdaIN aligns the combined high-frequency residual to the target statistics. Only the green blocks are trained.

with $q^{k+1} = P_H \bar{z}^{k+1}$, $K = 8$, $\beta = 2$, and $\eta = 0.1$. Because $\beta > 1$ the update extrapolates toward the matched statistics; applying it at every step makes alignment part of the dynamics rather than a post-process, and it supplies HH statistics without a learned head.

E. Metric-free Stopping

Training regresses a small latent field, needs no decoding, inversion, or adversary, and runs 52 steps per epoch at batch 96. After each epoch we probe a fixed batch of four latents at $t = 0.5$: ρ_e is the ratio of LL to HF gradient norm on the shared trunk. Training stops at the first epoch where the target-route gate grows and $\rho_e/\rho_{e-1} \leq 0.65$, i.e., when high-frequency learning takes over the trunk. On D5 this fires at epoch 4 (82.8 s), which is also the DINO-S peak of the 15-epoch curve; no external metric is used online.

IV. EXPERIMENTS

A. Setup

Benchmarks. *D5-512*: five WikiArt styles (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e) at 512 px, 3,600 training images each; 30 test sources per style are sent to all five targets (750 ordered requests, same-style rows kept). *P2A-256*: a 256 px grid over photo, Monet, Van Gogh, Cézanne, and Hayao. *R5-WikiArt*: five randomly drawn WikiArt families (Cubism, Expressionism, Pop Art, Romanticism, Symbolism) at 512 px.

Metrics. DINO-S (primary) is the maximum cosine between the DINOv2-small CLS feature [38] of the output and held-out references of the target style; CLIP-S is CLIP ViT-B/32 [39] similarity to the target-style prototype; DINO-C is DINOv2 similarity to the source; LPIPS uses AlexNet. IDT and TGT (a fixed held-out reference of the target style) are computed per board, and we inspect $M_{\text{style}}(y) > M_{\text{style}}(y_{\text{IDT}})$ and $\text{LPIPS}(y) < \text{LPIPS}(y_{\text{TGT}})$, $\text{DINO} - C(y) > \text{DINO} - C(y_{\text{TGT}})$ separately rather than combining them into one score.

Baselines and implementation. We compare training-free diffusion editors (SD-Turbo [40], StyleAligned, Z-STAR, StyleShot, StyleID), learned models (CUT, SaMST, SaMam, StyTR-2, AesPA-Net), the Seedream 4.5 API [41], and an analytic Latent-WCT, each at its released or recommended operating point. WEAVE uses AdamW (lr 2×10^{-4} , cosine), bf16, and one RTX 3060 (12 GB); training time excludes one-off latent caching. Inference time covers the 8-step ODE and VAE decoding for 750 images.

B. Main Results

Table I shows that the sandwich separates methods that single metrics conflate. SD-Turbo attains the best LPIPS (0.003) on D5 because it barely edits the input, and its CLIP-S does not exceed IDT. StyleAligned has the highest DINO-S on every board but crosses the TGT LPIPS reference on D5 and R5, i.e., it replaces rather than stylizes the content. SaMST on D5 and Latent-WCT on all boards stay below IDT in style; the latter shows that wavelet statistics alone do not replace learned transport.

On D5, WEAVE clears IDT on both style measures (DINO-S 0.4918 vs. 0.419, CLIP-S 0.7128 vs. 0.693) while keeping LPIPS at 0.260. Among trained methods that clear IDT on both style metrics, it has the lowest LPIPS; SaMam’s lower LPIPS (0.243) comes with CLIP-S below IDT. Compared with Seedream 4.5, WEAVE reaches higher DINO-S at roughly half the LPIPS. On P2A, WEAVE stays inside the bracket with the second-best DINO-C; on R5 it improves DINO-S over IDT, CUT, and SaMam with the second-best LPIPS. The stronger D5-512 profile is consistent with native high-frequency coefficients preserving detail that is compressed at 256 px, although P2A also differs in data. WEAVE has the fewest trainable parameters and the shortest training of all learned methods (1.4 min vs. 39.5–1440 min) and needs 168 ms per image at inference.

Qualitative comparison. Fig. 4 shows the same pattern visually: editors that score high on style redraw the scene, compact

TABLE I

MAIN COMPARISON (D-S: DINO-S, C-S: CLIP-S, LP: LPIPS, D-C: DINO-C). EXCLUDING IDT AND TGT, THE BEST ENTRY PER COLUMN IS **BOLD** AND THE SECOND UNDERLINED. †: STYLE NOT ABOVE IDT; ‡: CONTENT BEYOND TGT. PARAMS IN MILLIONS WITH FROZEN PARAMETERS AS SUPERScript; PARAMS AND TRAIN RANK TRAINED METHODS. WEAVE USES ITS SELECTED D5 CHECKPOINT ON D5 AND DATASET-SPECIFIC CHECKPOINTS ON P2A/R5; COST REFERS TO THE D5 MODEL. TIMES ON ONE RTX 3060; *: AUTHOR REPORT ON FOUR GPUS [13]; ~3 h: Z-STAR ESTIMATED FROM STEP-SCALED RUNS (FULL 512 PX RUNS EXCEED 12 GB).

Method	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer
	D-S†	C-S†	LP↓	D-C†	D-S†	C-S†	LP↓	D-C†	D-S†	C-S†	LP↓	D-C†	(M)	(min)	(750)
IDT (identity)	0.419	0.693	0.000	1.000	0.416	0.663	0.000	1.000	0.433	0.731	0.000	1.000	0	free	0 s
TGT (target style)	1.000	0.863	0.776	0.215	1.000	0.835	0.800	0.169	1.000	0.796	0.747	0.230	0	free	0 s
SD-Turbo	0.484	0.693 [†]	0.003	0.922	0.341 [†]	0.674	0.603	0.279	0.505	0.767	0.449	0.538	0 ⁺¹¹⁰⁰	free	5 m
StyleAligned	0.675	0.780	0.869 [‡]	0.239	0.612	0.768	0.786	0.310	0.649	0.824	0.829 [‡]	0.315	0 ⁺¹¹⁰⁰	free	77 m
Z-STAR	0.449	0.784	0.347	0.549	0.498	0.786	0.332	0.552	0.514	<u>0.822</u>	0.384	0.526	0 ⁺¹¹⁰⁰	free	~3 h
StyleShot	<u>0.563</u>	<u>0.787</u>	0.765	0.377	<u>0.552</u>	<u>0.774</u>	0.713	0.335	0.484	0.787	0.795 [‡]	0.380	0 ⁺²⁴⁰⁰	free	5.1 h
StyleID	0.548	0.822	0.552	0.416	<u>0.256</u> [†]	<u>0.648</u> [†]	0.691	0.147 [‡]	<u>0.551</u>	0.815	0.493	0.385	0 ⁺¹¹⁰⁰	free	63 m
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710 [†]	0.620	<u>0.795</u>	<u>7.0</u>	322.6	5 m
SaMST	0.271 [†]	0.618 [†]	0.749	0.145 [‡]	0.501	0.709	0.398	0.838	0.461	0.667 [†]	0.612	0.615	8.3	<u>39.5</u>	10 m
SaMam	0.477	0.582 [†]	<u>0.243</u>	<u>0.812</u>	0.505	0.677	0.205	0.870	0.503	0.712 [†]	0.227	0.925	8.8	436.0	17.6 m
StyTR-2	0.501	0.709	0.596	0.747	0.515	0.722	0.495	0.555	0.533	0.759	0.553	0.737	35.4 ^{+12.9}	~1440*	38 m
AesPA-Net	0.520	0.724	0.548	0.601	0.498	0.744	0.571	0.522	0.545	0.782	0.409	0.728	11.3 ^{+20.0}	–	23 m
Seedream 4.5	0.486	0.720	0.477	0.739	0.517	0.752	<u>0.227</u>	0.785	0.504	0.742	0.486	0.725	–	API	–
Latent-WCT	0.362 [†]	0.673 [†]	0.441	0.559	0.338 [†]	0.738	0.585	0.386	0.414 [†]	0.710 [†]	0.444	0.553	0	free	18 s
WEAVE (ours)	0.492	0.713	0.260	0.810	0.480	0.668	0.312	<u>0.861</u>	0.523	0.775	<u>0.289</u>	0.772	1.04 ⁺⁸⁴	1.4	<u>126 s</u>

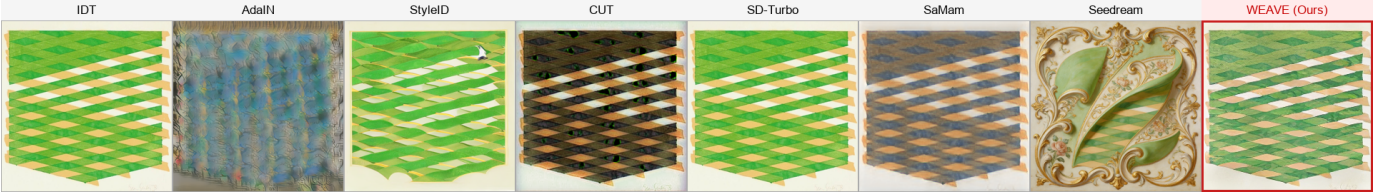


Fig. 4. Extreme cross-domain request (Minimalism → Rococo). Diffusion editors substantially alter source structure, and several lightweight or classical outputs remain close to identity. WEAVE retains the source layout while adding the painterly target appearance.

baselines add color casts but little texture, and WEAVE transfers brushwork onto the preserved source layout.

C. Mechanistic Ablations

Table II changes one component at a time on the same 750 requests. Inference-time controls reuse the selected checkpoint; retrained variants use seed 42 and are reported at their best-DINO-S epoch. Removing stepwise AdaIN lowers DINO-S by 0.0097 and CLIP-S by 0.0123 and moves the output toward the source. Removing the high-frequency conditioning lowers DINO-S and raises LPIPS. Replacing the source-aligned endpoint with the direct target has the largest content cost (LPIPS 0.2595→0.3138), and its peak arrives nine epochs later. Raising λ_{LL} from 0.3 to 1.0 changes peak DINO-S by less than 0.001 but lowers DINO-C to 0.787, so the method does not hinge on a finely tuned weight. A learned HH head gives a small style gain for a small content cost, so we keep the simpler model.

D. Robustness and Transfer

Seeds and stopping. The stopping rule selects epochs 4, 4, and 3 for seeds 42, 7, and 123, giving DINO-S 0.4918, 0.4910,

TABLE II
COMPONENT ABLATION ON D5-512 (750 REQUESTS).

Variant	DINO-S†	CLIP-S†	LPIPS↓	DINO-C†
WEAVE (epoch 4)	0.4918	0.7128	0.2595	0.8102
w/o stepwise AdaIN ($\beta=0$)	0.4821	0.7005	0.2258	0.8455
w/o HF conditioning	0.4825	0.7157	0.2837	0.8125
direct endpoint ($z_1^\dagger=z_s$)	0.4894	0.7172	0.3138	0.7966
$\lambda_{LL}=1.0$	0.4910	0.7180	0.2626	0.7874
learned HH head	0.4930	0.7164	0.2670	0.8061

and 0.4862 (LPIPS 0.260, 0.267, 0.255). For seeds 42 and 7, where full 15-epoch curves exist, the selected epoch coincides with the DINO-S peak.

Reference pools. Resampling a size- m reference pool per target style 1,000 times, the board-mean DINO-S margin of WEAVE over IDT is positive in every draw: 0.030 (95% interval [0.020, 0.039]) for $m=8$ and 0.032 ([0.023, 0.038]) for $m=16$. About 69% of individual requests improve, so the claim is board-level rather than per image.

Unseen families. Applied without retraining to five held-out WikiArt families (Table III, top), the D5 checkpoint gives the

TABLE III
TRANSFER BEYOND THE TRAINING SETTING. TOP: ZERO-SHOT ON FIVE
UNSEEN WIKIART FAMILIES (750 PAIRS, D5 CHECKPOINTS). BOTTOM:
HAAR ALIGNMENT INSIDE A FROZEN SD1.5 EDITOR (600 PAIRS).

Method	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$
SaMST	0.3913	0.7485	0.6213	0.2680
SaMam	0.5466	0.7577	0.3990	0.7369
WEAVE	0.5313	0.7742	0.2802	0.7587
SD1.5 img2img + Haar alignment	0.3974	0.7221	0.3591	0.6589
	0.4080	0.7366	0.3630	0.6531

best CLIP-S, LPIPS, and DINO-C; SaMam is 0.015 higher on DINO-S at a much higher LPIPS.

Plug-in for a frozen editor. We add the same Haar-domain alignment (LH/HL/HH, scale 0.8, LL untouched) to a frozen SD1.5 image-to-image pipeline (empty prompt, strength 0.5, 20 DDIM steps). On 600 cross-style D5 pairs, DINO-S and CLIP-S rise significantly (paired t -test, $p < 10^{-15}$) at a small LPIPS change (Table III, bottom), for 4.9 ms per image and no weight update.

V. LIMITATIONS AND CONCLUSION

WEAVE is evaluated with one SD1.5 VAE and a single Haar level; styles dominated by global palette or layout, or expressed across several scales, may need a deeper decomposition. Haar separates latent coordinates, not semantic factors, and the nonlinear decoder can couple them. The stopping rule is validated on one architecture and three seeds, and IDT/TGT are board-specific references measured with imperfect perceptual proxies. Baselines use their recommended operating points rather than a compute-matched search.

The IDT-TGT sandwich gives style and content metrics an explicit direction: IDT flags methods stuck at the source, TGT flags those that abandon it. WEAVE resolves the underlying conflict in frequency coordinates: a source-aligned target limits structural motion, band-weighted flow learns organized transport, and stepwise AdaIN supplies high-frequency statistics along the trajectory, with 1.04M trainable parameters and 1.4 minutes of training. Code, checkpoints, and the 750-request protocol will be released; the supplementary material details the protocol, per-epoch curves, and additional controls.

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